```
# data dividation
# workflow of ml
# data ingestion==>data cleaning and preprocessing -->based on the target co
# we have to select ml model==>model tranining ==>model prediction ==>model
# evaluation
```

```
# df===>x(input),y(target)
# x==>x_train,x_test
# y===>y_train,y_test
# model traning(x-train,y_train)
# y_pred=model prediction(x_test)
# model performence(y_test,y_pred)
```

```
# 1+1=2
# 1+2=3
# 1+3=4
# 1+4=5
# x=[1+1,1+2,1+3,1+4]
# y=[2,3,4,5]
# x_train=[1+1,1+2,1+3]
# y_train=[2,3,4]
# x_test=[1+4]
# y_test=[5]
```

```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('/content/placement.csv')
df.head()
```

	cgpa	resume_score	placed
0	8.14	6.52	1
1	6.17	5.17	0
2	8.27	8.86	1
3	6.88	7.27	1
4	7.52	7.30	1

Next steps: Generate code with df | New interactive sheet

```
df.shape
```

```
(100, 3)
```

```
x=df.drop(columns=['placed'])
#x
y=df['placed']
#y
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_stat
```

```
print('total shape of df:',df.shape)
print('------****-----')
print('total input data shape',x.shape)
print('x_train data shape',x_train.shape)
print('x_test data shape',x_test.shape)
print('----')
print('total output data shape',y.shape)
print('y_train data shape',y_train.shape)
print('y_test data shape',y_test.shape)
```

```
total shape of df: (100, 3)
------****-----
total input data shape (100, 2)
x_train data shape (80, 2)
x_test data shape (20, 2)
----
total output data shape (100,)
y_train data shape (80,)
y_test data shape (20,)
```

```
# df(100,3)=x(100,2)+y(100,1)
# x(100,2)=x_train(80,2)+x_test(20,2)
# y(100,1)=y_train(80,1)+y_test(20,1)
```

```
# task=covid dataset convert into categorical to numerical,missing data hand
# df=x,y,x_train,x_test,y_train,y_test
```

```
df1=pd.read_csv('/content/covid_toy.csv')
df1.head(2)
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	No
1	27	Male	100.0	Mild	Delhi	Yes

Next steps: Generate code with df1 New interactive sheet

```
# Agar pure dataset ke categorical columns convert karne hain
# for col in df.select_dtypes(include='object').columns:
#     df[col] = df[col].map({
#         value: i for i, value in enumerate(df[col].unique())
#     })
```

```
df1.select_dtypes(include='object').columns
```

```
Index(['gender', 'cough', 'city', 'has_covid'], dtype='object')
```

```
df1['has_covid'].value_counts()
```

has_covid	count
No	55
Yes	45

dtype: int64

```
df1['gender'] = df1['gender'].map({'Male': 0, 'Female':

df1['cough'] = df1['cough'].map({'Mild': 0, 'Strong': 1

df1['city'] = df1['city'].map({'Kolkata': 0, 'Delhi': 1

df1['has_covid'] = df1['has_covid'].map({'No': 0, 'Yes'
```

```
df1.select_dtypes(include='int').columns
```

```
Index(['age'], dtype='object')
```

```
df1['fever']=df1['fever'].fillna(df1['fever'].mean())
```

```
df1['fever'].mean()
```

```
np.float64(100.84444444444445)
```

```
x=df1.drop(columns=['has_covid'])
y=df1['has_covid']
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_stat
```

```
print('total shape of df:',df1.shape)
print('------****-----')
print('total input data shape',x.shape)
```

```
print('x_train data shape',x_train.shape)
print('x_test data shape',x_test.shape)
print('----')
print('total output data shape',y.shape)
print('y_train data shape',y_train.shape)
print('y_test data shape',y_test.shape)
```

```
total shape of df: (100, 6)
------****------
total input data shape (100, 5)
x_train data shape (80, 5)
x_test data shape (20, 5)
----
total output data shape (100,)
y_train data shape (80,)
y_test data shape (20,)
```